DSA Basics (30-40 Hours)

Module 1: Introduction to Data Structures and Algorithms

- **Duration**: 2-3 Hours
  - Importance of DSA
  - Time and Space Complexity (Big O, Big Theta, Big Omega)
  - Introduction to Recursion
  - Understanding Problem-Solving Techniques

Module 2: Arrays

- **Duration**: 5-6 Hours
  - Introduction to Arrays
  - Types of Arrays (1D, 2D, Multi-dimensional)
  - Array Operations (Insertion, Deletion, Traversal)
  - Searching in Arrays (Linear Search, Binary Search)
  - Sorting Algorithms (Bubble Sort, Selection Sort, Insertion Sort)
  - Applications of Arrays (Matrix Multiplication, etc.)

Module 3: Linked Lists

- **Duration**: 5-6 Hours
  - Introduction to Linked Lists
  - Types: Singly Linked List, Doubly Linked List, Circular Linked List
  - Operations (Insertion, Deletion, Traversal)
  - Reversing a Linked List
  - Detecting and Removing Cycles in Linked Lists
  - Applications of Linked Lists

Module 4: Stacks and Queues

- **Duration**: 4-5 Hours
  - Introduction to Stacks (LIFO Principle)
  - Stack Operations (Push, Pop, Peek)
  - Applications of Stacks (Expression Evaluation, Backtracking, etc.)
  - Introduction to Queues (FIFO Principle)
  - Queue Operations (Enqueue, Dequeue, Peek)
  - Types of Queues (Simple Queue, Circular Queue, Priority Queue, Deque)
  - Applications of Queues (Scheduling, Buffering, etc.)

Module 5: Trees

- **Duration**: 5-6 Hours
  - Introduction to Trees and Terminology (Root, Leaf, Node, etc.)
  - Binary Trees: Types, Operations, and Traversal (Inorder, Preorder, Postorder)
  - Binary Search Tree (BST): Insertion, Deletion, Search
  - Balanced Trees: AVL Trees, Red-Black Trees
  - Applications of Trees (Expression Trees, Huffman Encoding)

Module 6: Heaps

- **Duration**: 3-4 Hours
  - Introduction to Heaps
  - Types: Min Heap, Max Heap

    - Operations on Heaps (Insertion, Deletion, Heapify)
    - Heap Sort Algorithm
    - Priority Queues using Heaps

Module 7: Hashing

- **Duration**: 3-4 Hours
    - Introduction to Hashing
    - Hash Functions and Collision Resolution Techniques (Chaining, Open Addressing)
    - Applications of Hashing (Dictionary, Caching, etc.)
    - Load Factor and Rehashing

Module 8: Graphs

- **Duration**: 5-6 Hours
    - Introduction to Graphs and Terminology (Vertices, Edges, etc.)
    - Types of Graphs (Directed, Undirected, Weighted, Unweighted)
    - Graph Representation (Adjacency Matrix, Adjacency List)
    - Graph Traversal Techniques (BFS, DFS)
    - Shortest Path Algorithms (Dijkstra's, Bellman-Ford)
    - Minimum Spanning Tree (Kruskal's, Prim's)
    - Applications of Graphs (Social Networks, Pathfinding, etc.)

DSA Advanced (40-50 Hours)

Module 9: Advanced Sorting and Searching

- **Duration**: 4-5 Hours
    - Quick Sort, Merge Sort, and Heap Sort
    - Counting Sort, Radix Sort, and Bucket Sort
    - Advanced Searching Techniques (Ternary Search, Exponential Search)

Module 10: Dynamic Programming (DP)

- **Duration**: 6-7 Hours
    - Introduction to Dynamic Programming
    - Memoization vs Tabulation
    - Classical DP Problems: Knapsack, Longest Common Subsequence, Longest Increasing Subsequence
    - Optimal Substructure and Overlapping Subproblems
    - Advanced DP Problems: Matrix Chain Multiplication, Partition Problems

Module 11: Greedy Algorithms

- **Duration**: 4-5 Hours
    - Introduction to Greedy Methodology
    - Classical Greedy Problems: Activity Selection, Fractional Knapsack, Job Sequencing
    - Huffman Coding
    - Applications of Greedy Algorithms

Module 12: Backtracking

- **Duration**: 4-5 Hours
    - Introduction to Backtracking
    - Solving Problems Using Backtracking (N-Queens, Sudoku Solver, Rat in a Maze)

    - Pruning Techniques and Optimizations

Module 13: Advanced Graph Algorithms

- **Duration**: 6-7 Hours
    - Topological Sorting
    - Strongly Connected Components (Kosaraju's Algorithm)
    - Graph Coloring
    - Network Flow (Ford-Fulkerson, Edmonds-Karp)
    - Applications in Network Design, Circuit Design, etc.

Module 14: Segment Trees and Binary Indexed Trees (BIT)

- **Duration**: 4-5 Hours
    - Introduction to Segment Trees
    - Range Queries and Point Updates
    - Lazy Propagation in Segment Trees
    - Binary Indexed Tree (Fenwick Tree) for Range Queries
    - Applications in Competitive Programming

Module 15: Trie and Suffix Trees

- **Duration**: 4-5 Hours
    - Introduction to Tries
    - Implementing Tries (Insert, Search, Delete)
    - Applications of Tries (Prefix Matching, Auto-Complete, etc.)
    - Introduction to Suffix Trees
    - Applications in Pattern Matching

Module 16: Advanced Data Structures

- **Duration**: 4-5 Hours
    - Disjoint Set (Union-Find) with Path Compression and Union by Rank
    - Bloom Filters
    - B-Trees and B+ Trees
    - Fibonacci Heaps
    - Applications in Databases, File Systems, etc.

Module 17: Final Project

- **Duration**: 6-8 Hours
    - Choose a Complex Problem or System to Solve
    - Integrate Multiple DSA Concepts
    - Optimize for Performance and Scalability
    - Code Review and Final Presentation

Conclusion and Review

- **Duration**: 2-3 Hours
    - Review of Advanced Concepts
    - Practice with Competitive Programming Problems
    - Tips for Interviews and Further Learning